Machine Learning

06.05.26

Gaussian Naive Bayes :

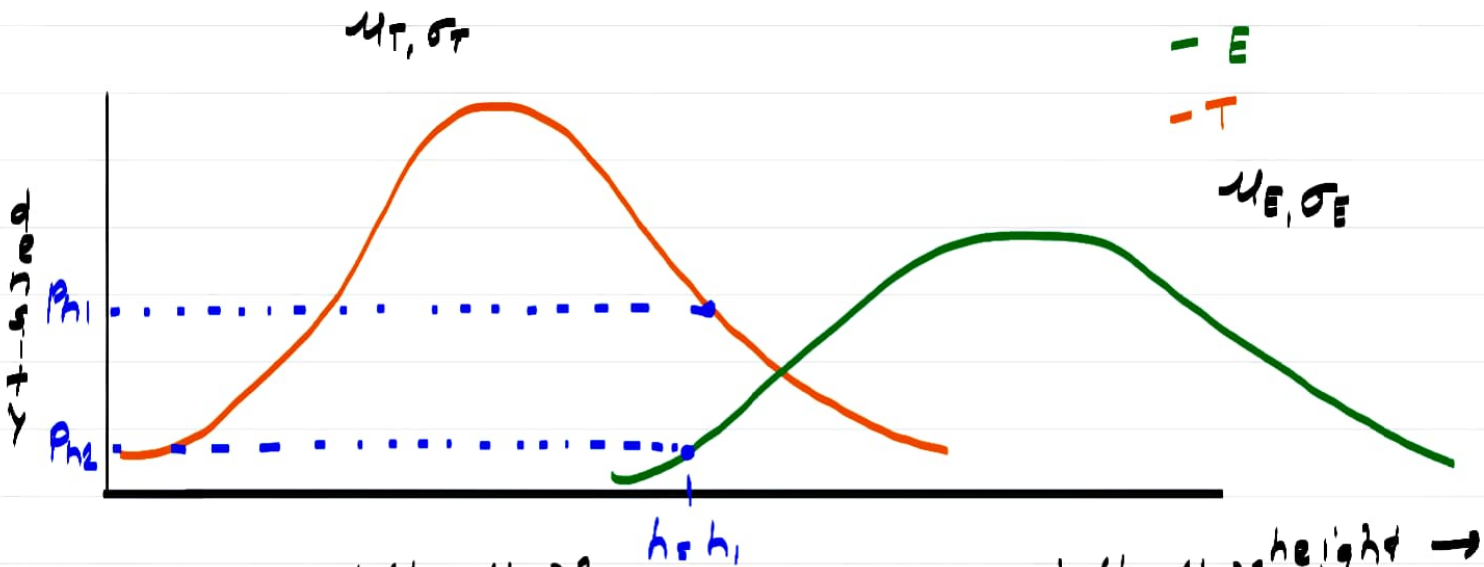
- used when features are continuous variables
- assumes that features are normaly distributed for all target classes
- features need to have conditional independence

eg.

height	weight	class (T/E)
5.5	1.2	T
6	4.6	T
4.2	1.8	E
3.4	4.7	E
⋮	⋮	⋮

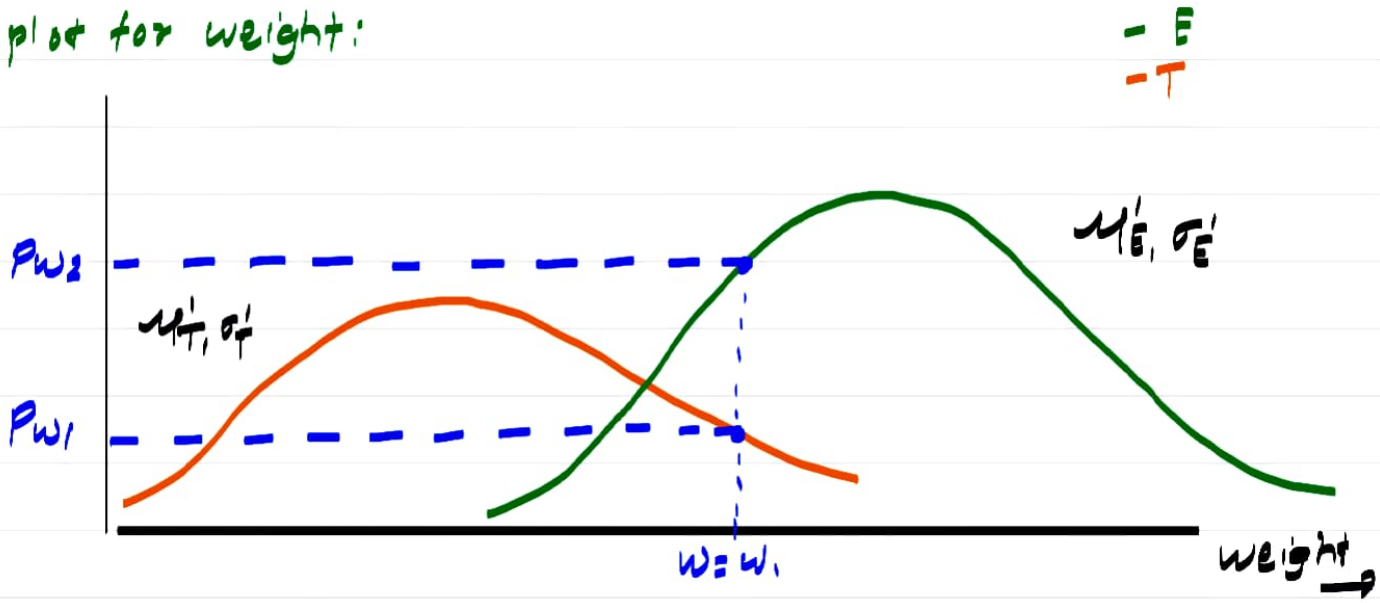
10000 - total
6000 - E + 4000 T

KDE plot for height - train data



$$P_{h_1} = \frac{1}{\sqrt{2\pi}\,\sigma_T} \times e^{-\frac{1}{2}\left(\frac{h_1-\mu_T}{\sigma_T}\right)^2} \qquad P_{h_2} = \frac{1}{2\pi\sigma_E} \times e^{-\frac{1}{2}\left(\frac{h_1-\mu_E}{\sigma_E}\right)^2}$$

KDE plot for weight:



$$P_{w_1} = \frac{1}{\sqrt{2\pi}\,\sigma'_T} \times e^{-1/2\left(\frac{w_1 - \mu'_T}{\sigma'_T}\right)^2} \qquad P_{w_2} = \frac{1}{\sqrt{2\pi}\,\sigma'_E} \times e^{-1/2\left(\frac{w_2 - \mu'_E}{\sigma'_E}\right)^2}$$

$h = h_1$, $w = w_1$ $\longrightarrow$ $\boxed{\text{gaussian NB clf}}$ $\longrightarrow$ T/E

$$P(T \mid h = h_1 \text{ and } w = w_1) \propto P(h = h_1 \text{ and } w = w_1 \mid T) \cdot P(T)$$
$$\propto P(h = h_1 \cap w = w_1 \mid T) \cdot P(T)$$
$$\propto P(h = h_1 \mid T) \cdot P(w = w_1 \mid T) \cdot P(T)$$

$$P(h = h_1 \mid T) = P_{h_1} = \frac{1}{\sqrt{2\pi}\,\sigma_T} \times e^{-1/2\left(\frac{h_1 - \mu_T}{\sigma_T}\right)^2}$$

$$P(w = w_1 \mid T) = P_{w_1} = \frac{1}{\sqrt{2\pi}\,\sigma'_T} \times e^{-1/2\left(\frac{w_1 - \mu'_T}{\sigma'_T}\right)^2}$$

$$P(T) = \frac{4000}{10000} = \frac{4}{10}$$

$$P(E \mid h=h_1 \text{ and } w=w_1) \propto P(h=h_1 \text{ and } w=w_1 \mid E) \cdot P(E)$$
$$\propto P(h=h_1 \cap w=w_1 \mid E) \cdot P(E)$$
$$\propto P(h=h_1 \mid E) \cdot P(w=w_1 \mid E) \cdot P(E)$$

$$P(h=h_1 \mid E) = \frac{1}{\sqrt{2\pi}\,\sigma_E} e^{-1/2\left(\frac{h_1-\mu_E}{\sigma_E}\right)^2}$$

$$P(w=w_1 \mid E) = \frac{1}{\sqrt{2\pi}\,\sigma_E'} \times e^{-1/2\left(\frac{w_1-\mu_E'}{\sigma_E'}\right)^2}$$

$$P(E) = \frac{6000}{10000} = \frac{6}{10}$$

find max among

$P(T \mid h=h_1 \text{ and } w=w_1)$ and $P(E \mid h=h_1 \text{ and } w=w_1)$

lets say $P(T \mid h=h_1 \text{ and } w=w_1) > P(E \mid h=h_1 \text{ and } w=w_1)$
then

$$\begin{matrix} h=h_1 \\ w=w_1 \end{matrix} \longrightarrow \boxed{\begin{matrix}\text{gaussian} \\ \text{NB clf}\end{matrix}} \longrightarrow T$$